

External Auditors' Judgment and Decision Making: An Audit Process Task Analysis*

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Received on November 13, 2017; editorial decision completed on July 17, 2018

ABSTRACT

This manuscript reviews the external auditor judgment and decision making (JDM) literature from the past decade by specific tasks in the auditing process. Decomposing the literature review by audit task was chosen to enable audit and behavioral researchers and graduate students to identify the trends of JDM research subsequent to Nelson and Tan's (2005) literature review. In addition to identifying gaps/omissions in the current literature, suggestions for future research are also advanced based on forthcoming transformations in the auditing institutional environment.

Keywords Audit JDM literature review; Auditor decision making; Auditor judgments

JUGEMENT ET PRISE DE DÉCISIONS DES AUDITEURS EXTERNES : UNE ANALYSE DES TÂCHES DU PROCESSUS D'AUDIT

RÉSUMÉ

Les auteurs passent en revue les études sur le jugement et la prise de décisions des auditeurs externes publiées au cours de la dernière décennie, en fonction de tâches précises du processus d'audit. Leur choix de segmenter leur analyse documentaire selon les tâches d'audit vise à permettre aux chercheurs et aux étudiants de cycle supérieur en études comportementales de définir les tendances de la recherche sur le jugement et la prise de décisions postérieure à l'analyse documentaire de Nelson et Tan (2005). En plus de relever les lacunes et les oublis de la documentation récente, les auteurs suggèrent des pistes de recherche pour l'avenir, en tenant compte des transformations prévisibles de l'environnement institutionnel de l'audit.

Mots clés : Jugements des auditeurs, Prise de décisions des auditeurs, Revue de la documentation sur le jugement et la prise de décisions en audit

This manuscript reviews the external auditing judgment and decision making (JDM) literature over the past decade by auditing task. Auditing JDM research is “research that uses a psychological lens to understand, evaluate, and improve judgments, decisions, or

*Accepted by Pascale Lapointe-Antunes. The authors thank CPA Ontario and the CPA Ontario Centre for Capital Markets and Behavioral Decision Making for financial support. The research assistance from Sara Wick and Jaspreet Malhi is greatly appreciated. Regan Schmidt acknowledges the Edwards Research Scholar financial support from the Edwards School of Business.

choices in an auditing setting” (Nelson and Tan, 2005: 41). As a result, the majority of these studies use an experimental methodology, however, as applicable, other research methods are used and included in this review, such as survey research. We have chosen to decompose the review by auditing task (see Appendix) based on calls in prior research to better understand the process of auditing (Felix and Kinney, 1982; Solomon and Shields, 1995; Bonner and Pennington, 1991; Abdolmohammadi, 1999; Kinney, 2005; Messier, 2010). Because this review has been structured by audit task, our review focuses on external auditors’ JDM in their audit of financial statements. Reviews of other participants’ JDM (e.g., internal auditors (IAs), management, and audit committees) are excluded from this review and have been reviewed elsewhere (e.g., Roussy and Perron, 2018; Gramling, Maletta, Schneider, and Church, 2004; Hass, Abdolmohammadi, and Burnaby, 2006; Allegrini, D’Onza, Paape, Melville, and Sarens, 2006; Cooper, Leung, and Wong, 2006; Carcello, Hermanson, and Ye, 2011; Trotman, Tan, and Ang 2011). By decomposing the review by auditing task, we facilitate audit researchers and graduate students’ understanding of the current pulse of auditing JDM research and identify potential gaps and areas for future research.

Our manuscript contains four sections. The first section discusses the auditing process and tasks that serve as the foundation for our review. As outlined in the Appendix, subsections commence with ethical requirements, professional skepticism, quality control, and association (e.g., auditor objectivity) and then proceed to planning judgments (e.g., risk assessments), execution and evaluation judgments (e.g., internal control testing), and reporting judgments (e.g., going concern evaluation). The second section discusses the method for selecting papers and conducting the literature review. Our review is restricted to manuscripts focused on external auditor JDM published in the following five journals: *The Accounting Review*, *Journal of Accounting Research*, *Contemporary Accounting Research*, *Accounting, Organizations and Society*, and *Auditing: A Journal of Practice & Theory*. We selected these journals as they include top general interest accounting journals that publish JDM research as well as the most significant auditing journal. This also is consistent with the journals covered by the Nelson and Tan (2005) review. The third section reviews the auditor JDM research by auditing task in the sequential flow of the audit process. The review summarizes the previous findings from Nelson and Tan in the applicable stage of the audit process,¹ identifies relevant Canadian Auditing Standards (CAS), reviews the past decade of literature and summarizes how the literature has been advanced since Nelson and Tan’s review (where applicable). Each audit process subsection concludes with suggestions for future research based on transformations that are currently occurring in the auditing institutional context. The fourth section of the manuscript provides a conclusion and discusses the manuscript’s limitations.

1. Nelson and Tan (2005) identify only four specific audit tasks (risk assessments/audit planning, analytical procedures, correction decisions, and opinion modification). Applicable sections of the Nelson and Tan review contained in its other sections (i.e., “the auditor” and “interpersonal interactions” sections) are summarized if applicable given the task context of the current manuscript.

AUDITING PROCESS AND TASKS

The Appendix outlines the four broad stages of the audit process as (i) ethical requirements, professional skepticism, quality control and association, (ii) planning judgments, (iii) execution and evaluation judgments, and (iv) reporting judgments. Within each stage of the audit process, specific auditing tasks are identified (Felix and Kinney, 1982; Solomon and Shields, 1995; Bonner and Pennington, 1991; Abdolmohammadi, 1999; Messier, 2010). These audit tasks serve as the foundation for our review. We note that some auditing tasks are pervasive throughout the entire audit process (e.g., professional skepticism) and other auditing tasks may be categorized in more than one stage of the audit process (e.g., analytical procedures), thus, we situated auditing tasks in the most appropriate stage of the audit process, yielding nine broad auditing tasks, as illustrated in the Appendix. Further, the auditing tasks listed in the appendix are not intended to be comprehensive as many auditing tasks have not been researched in the period under review.

METHOD

In this section, we describe our approach to systematically identify and analyze the auditing JDM literature that is included in the scope of this article. In determining the years and journals to review, we chose to remain consistent with the auditing JDM review conducted by Nelson and Tan (2005). Accordingly, given their review included published manuscripts and working papers up to and including 2005, our review commenced with papers published from 2006 to 2016. Notably, this provides a small degree of overlap as some papers included in Nelson and Tan (2005) as working papers or forthcoming articles were then published in the period of our review (e.g., Shankar and Tan, 2006). Further, again to remain consistent with Nelson and Tan, this review is restricted to manuscripts focused on external auditor JDM published in *The Accounting Review*; *Journal of Accounting Research*; *Contemporary Accounting Research*; *Accounting, Organizations and Society*; and *Auditing: A Journal of Practice & Theory*.

We followed a four-step process to compile the list of relevant literature for our review. We first compiled a list of audit tasks, as identified above. Then, for each journal, the table of contents and abstracts of each article in each journal for the period under review were read to identify auditor JDM papers. Once identified, all auditor JDM and potential auditor JDM papers were initially coded by task based on the abstract. The authors then read each paper to ensure appropriate task classification. If an author disagreed with the classification, the authors discussed the paper and its classification. Given the sequential nature of the auditing process, a particular manuscript may address several auditing judgments and tasks. In such circumstances, the authors were compelled to use professional judgment to categorize the manuscript into the most fitting auditing task for the purpose of the review. In addition, a small number of manuscripts were not amenable to audit task allocation and therefore were not included in the review. Finally, although every effort was made to ensure all manuscripts that merit inclusion in the review were identified, the authors make no assertion regarding completeness. Likewise, many published manuscripts exist in journals beyond the five journals listed above that further advance our understanding of external auditor JDM.

LITERATURE REVIEW BY AUDIT TASK

Ethical Requirements, Professional Skepticism, Quality Control, and Association Judgments

Judgments made during this part of the audit process are pervasive and impact all subsequent decisions (e.g., client acceptance as an element of quality control). In regard to prior research in this area, Nelson and Tan's (2005) review addressed individual auditor characteristics, such as knowledge and expertise, which are essential elements of the quality control of an audit engagement (e.g., human resources). Their review highlighted how auditor expertise is not equivalent to experience. With respect to auditor knowledge (e.g., tacit managerial knowledge), their review outlined how auditors, in particular audit seniors rather than audit managers, have been found to be overconfident in assessing their subordinates' technical knowledge. Lastly, auditor individual characteristics saw a resurgence in research during the time period of Nelson and Tan's review — for example, how problem-solving ability impacts auditor performance. To review the literature, this stage of the audit process is decomposed into (i) auditor ethical requirements and professional skepticism and (ii) firm quality control tasks.

Auditor Ethical Requirements and Professional Skepticism

CAS 200.14 requires auditors to comply with ethical requirements in the audit of financial statements.² Ethical requirements comprise independence and other ethical requirements included in the code of conduct for the accounting profession (CAS 200.CA16-CA17). Although the jurisdiction for codes of conduct and its enforcement reside at a provincial level in Canada (Mescall, Phillips, and Schmidt 2017), fundamental principles embodied in the code of conduct for the Canadian accounting profession include objectivity, integrity, professional competence and due care, confidentiality, and professional behavior (CAS 200.CA17; Chartered Professional Accountants of Ontario, 2016). Although objectivity is a topic frequently examined in the auditing literature, the remaining ethical principles tend to be less frequently examined as discussed below.

Objectivity. Recent literature in the objectivity area has focused on familiarity threats to auditors' independence. Bamber and Iyer (2007) posit that auditors' familiarity with clients may be both a threat to independence yet, to some extent, also necessary for an effective and efficient audit. In the context of a financial reporting disagreement regarding materiality of unrecorded liabilities, their study measured the extent to which auditors identify with both the client and the profession. Bamber and Iyer's results

2. "CASs result from the adoption of International Standards on Auditing (ISAs), developed and issued by the IAASB, that are consistent with the *International Framework for Assurance Engagements*. The Auditing and Assurance Standards Board (AASB) in Canada, may make amendments to ISAs that affect how a practitioner performs an audit of financial statements or change the effective date of application of ISAs, in accordance with criteria that the AASB has developed for ensuring that changes to standards are made on an appropriate and consistent basis." (Chartered Professional Accountants Canada, 2017, paragraphs 19 and 20).

document that auditors who identify with the client are more likely to accept the client-preferred accounting treatment (i.e., to not record the liabilities). However, auditors who identify with the profession and/or have greater audit experience are less likely to accept the client-preferred accounting treatment. Stefaniak, Houston, and Cornell (2012) extend Bamber and Iyer's (2007) research to examine whether the influence of client identification generalizes to a context of internal control evaluations (i.e., IT access control deficiencies). Stefaniak, Houston, and Cornell (2012) conceptually replicate Bamber and Iyer's (2007) results, such that auditors that identify with clients assess the controls relatively more leniently. Bauer (2015) furthers this stream of literature by documenting that auditors with strong client identification assess going concern likelihood as higher and accept smaller inventory write-downs compared to auditors with weak client identification; however, when professional identity salience is heightened these differences across conditions are no longer statistically significant. Bauer also finds that heightened professional identity salience increases auditors' professional skepticism.

The extent to which auditors' provision of non-audit services undermines objectivity has been a subject of great debate in the literature. Joe and Vandervelde (2007) examine if providing non-audit services improves auditors' judgments as a result of knowledge transfers between tasks. Their results suggest that knowledge does transfer from the non-audit services to the audit services in a beneficial manner. Specifically, when auditors perform both the audit and non-audit tasks, auditors have elevated risk assessments compared to having one auditor perform the non-audit task and a second auditor perform the audit task, but has access to the non-audit task working papers.

Ethical judgments. Coram, Glavovic, Ng, and Woodliff (2008) examine how auditors react to different fieldwork infractions that would reduce audit quality (e.g., failure to test all sample items). They investigate whether the "moral intensity" differs across the infractions; for example, whether auditors perceive the infraction as being wrong. Results suggest that auditors believe the infractions are equally wrong and could lead to an incorrect audit opinion. However, auditors believed the infractions differed in regard to a supervisor's ability to detect the infraction, and the disciplinary consequences of these infractions. Therefore, ethical auditor behavior may be contingent on the specific issue.

Ethical judgments have also been examined from an affect framework. Johnson, Lowe, and Reckers (2016) question whether auditors' mood impacts their willingness to comply with a superior auditors' unethical directives across different ethical scenarios (e.g., eating time). In addition to documenting a high degree of compliance intentions overall, the results show that auditors' mood does impact their compliance intentions. In this study, mood was measured using the participant's responses to 38 mood/affect adjectives from the Modified Positive Affect-Negative Affect Scale solicited after reading the case information. Whereas high levels of arousal decrease auditors' willingness to comply, both high levels of fear and insignificance increase auditors' willingness to comply with the superiors' unethical directives.

Professional skepticism. CAS 200.15 requires auditors to maintain professional skepticism in the audit of financial statements. Quadackers, Groot, and Wright (2014)

examine whether the attitude of professional skepticism implies auditors are neutral, with no assumption of biased management reporting, or have presumptive doubt, assuming management is biased. Using the Hurtt Professional Skepticism Scale to measure neutrality and the inverse of the Rotter Interpersonal Trust Scale to measure presumptive doubt (i.e., a lack of trust), their results document that, across risk scenarios, the presumptive doubt measure is more predictive of skeptical planning and risk assessment than the neutrality measure.

Bowlin, Hobson, and Piercey (2015) question how the interaction of auditors' mental frames of management representations (either assuming potential dishonesty or potential honesty) and auditor rotation affects auditors' effort. Their results suggest auditor rotation increases audit effort when auditors have a potential honesty mindset. However, auditor rotation decreases audit effort when auditors have a potential dishonesty mindset. The authors suggest that, in regard to audit effort, encouraging a skeptical dishonesty mindset is more efficient than mandating auditor rotation.

With respect to impediments to professional skepticism, Brazel, Jackson, Schaefer, and Stewart (2016) examine whether the review process within the audit firm is negatively impacting the professional skepticism of auditors. That is, do superiors penalize subordinates on performance evaluations when subordinates appropriately exercise professionally skeptical behavior, but fail to identify a misstatement? Results suggest that deleterious performance review effects occur and that although consultation with the superior improved performance evaluations, such consultation did not alleviate the outcome effect. Accordingly, the auditing firm performance evaluation system may inadvertently be discouraging auditors from exercising professional skepticism.

Firm Quality Control

Canadian Standards on Quality Control (CSQC-1) and CAS 220 outline the elements of the system of quality control used by the auditing firm to ensure the firm and its personnel comply with the standards/requirements of the profession and the reports issued by the firm are appropriate. Elements of this system of quality control include, but are not limited to, policies and procedures on leadership, and firm human resources.

Firm leadership. Audit firm superiors' (e.g., audit partners) views and preferences have an impact on the behavior of other auditors in the firm. Peytcheva and Gillett (2011) document that auditors' initial judgments of the accounting treatment of a fixed asset expenditure (i.e., capitalize vs. expense) are subsequently changed to coincide with the audit partner's view after the view of the audit partner are revealed. The magnitude of the change is indistinguishable from audit partner views being known at the time the initial judgment is made. The presence of this bias to coincide with the audit partner view suggests that reported auditor views are malleable to be consistent with firm superiors, potentially undermining audit quality.

Leadership styles have been shown to influence how audit teams interact. Nelson, Proell, and Randel (2016) document that, in comparison to an audit manager with a

personal advancement focus, an audit manager with a team-oriented leadership focus increases subordinate auditors' comfort to raise audit issues. Furthermore, the audit team member's willingness to raise audit issues is further influenced by the supervising audit partners' concerns on the engagement. Specifically, if an audit partner is concerned about audit effectiveness (efficiency), audit team members are more willing to raise issues related to audit effectiveness (efficiency) as compared to issues related to audit efficiency (effectiveness).

Human resources. Auditors work in a unique institutional context which contains multiple accountabilities, workload pressures, staffing decisions, and training. The impact of this context on auditors' behavior has received increased attention in the literature. Agoglia, Hatfield, and Lambert (2015) examine how underreporting time on various clients is viewed by audit managers and audit partners. Their results suggest that, when the client is desirable (i.e., favorable auditor–client relations, non-hectic audit timing, experienced audit partner, etc.), audit managers evaluate underreporting audit seniors more favorably than accurate reporters. Further, audit managers tend to request underreporting auditors, rather than accurate reporters, on future audit engagements. In contrast, audit partners do not differentiate between under-reporters and accurate reporters in their performance evaluations or requests for the senior on future engagements.

Auditors often are required to perform audits under time constraints imposed by the firms and clients. Prior work indicates that time constraints have a negative impact on outcomes. Low and Tan (2011) investigate whether early or late warnings of impending time constraints and instructions to think outside the box when designing audit procedures improve auditors' performance under time constraints. Results show that auditors perform better when informed earlier about time constraints and this performance is enhanced even more when auditors are instructed to think outside of the box. To further examine workplace time pressures, Agoglia, Brazel, Hatfield, and Jackson (2010) examine how such pressures impact the working paper review process. Their survey results document that reviewers view electronic reviews as more convenient, however in-person reviews are perceived to be more effective. Using these survey results to inform their experimental design, Agoglia et al. (2010) question whether work-load pressure interacts with risk of material misstatement to influence reviewers' choice of review method. Results suggest that only when the risk of material misstatement is low, does workload pressure increase the likelihood that reviewers choose to conduct an electronic review. However, if the misstatement risk is high, reviewers prefer to conduct an in-person review, regardless of the work-load pressure.

Recognizing that an auditor may be held accountable to multiple superiors with different levels of clarity and pressure, Bagley (2010) examines how these accountability complexities might impact auditors' negative affect. In contrast to when auditors are held accountable to a single superior, multiple superiors increase the negative affect of auditors, as measured by the Positive and Negative Affect Schedule scale. Moreover, negative affect is found to negatively impact low complexity audit task performance (i.e., listing audit procedures), but not high complexity audit task performance (i.e., ratio analysis).

Another stream of literature has examined how auditors assess the competence of the audit staff, which is critical to the human resource element of quality control. Prior research has documented that auditors are overconfident and inaccurate in assessing the competence of their colleagues, potentially undermining audit quality. Han, Jamal and, Tan (2011) examine how both audit superiors (i.e., audit managers) and subordinates (i.e., audit seniors) assess each other's technical competence for different audit tasks. When assessing their managers, seniors are more overconfident in their competence assessment for difficult task contexts (i.e., interest rate swaps) than easier task contexts (i.e., error frequencies). However, the relationship reverses when managers assess the competence of the seniors, such that the managers are more overconfident in competence assessment for easier task contexts than difficult task contexts. Harding and Trotman (2009) provide a potential solution by examining how to improve auditor competence assessments through different types of feedback. The results indicate that when the assessing auditor is familiar (unfamiliar) with the colleague, individual-specific (average-group) outcome feedback effectively reduces overconfidence in competence assessments.

Related to auditor competence, the human resource element of firm quality control standards also addresses how audit firms train auditors to have adequate knowledge and expertise (CSQC-1.A31). Knowledge and expertise have a rich tradition of study in the auditing JDM literature, but appear to have decreased in prominence during recent years. Borthick, Curtis, and Sriram (2006) find that training auditors on internal controls (either by focusing on transaction flow or control objectives) influences their knowledge structures. Moreover, the knowledge structures of auditors are found to mediate the relationship between training and subsequent performance in an internal control task. Although one type of knowledge structure does not result in superior performance (measured by inferences of control strength), the results document that auditors with highly structured knowledge perform better when the task demands are consistent with their knowledge structures (e.g., when both the knowledge structure and task demands were in transaction flow), as compared to when knowledge structure and task demands are mismatched.

Summary of Ethical Requirements, Professional Skepticism, Quality Control and Association Judgments, and Suggestions for Future Research

Although Nelson and Tan (2005) did not specifically identify auditing tasks in this part of the audit process in their review, they did address auditor characteristics which the literature has continued to investigate over the past decade. Notably, the importance of objectivity to audit quality has been highlighted in numerous manuscripts. In addition, rather than focusing merely on the effects of professional skepticism, the literature has moved beyond to identify how professional skepticism may be inadvertently discouraged through firm review processes. Research has provided empirical evidence to suggest timing issues including time reporting and time constraints are influential in auditor decision making. Auditor knowledge, both in regard to individual development and assessing audit team members, continues to be a stream of interest. Finally, the malleability of auditors has also been emphasized as auditors are biased to be consistent with the views and concerns of superiors.

In regard to future research, clearly some elements of quality control (e.g., auditor acceptance and continuance decisions) and the impact of association standards have not been examined in the period under review from the auditor perspective, despite the evolution of such standards during this period. Further, auditor competencies and expertise will continue to be a fruitful area of research. Certainly, the demand for new assurance services beyond traditional financial statement audits will require audit researchers to provide insights into how to understand, assess and ultimately improve such services, notably in corporate social responsibility and sustainability (e.g., O'Dwyer, 2011; Huang and Watson, 2015; Cohen and Simnett, 2015). Focusing on financial statement auditing, opaque auditor independence issues, such as, social ties are in need of future research, especially given that recent archival research has documented detrimental effects on audit quality when social ties exist between auditors and the audit committee (He, Pittman, Rui, and Wu 2017).

Directly related to financial statement auditing and professional standards on association, companies are increasingly using technology in the form of websites (Ettredge, Richardson, and Scholz 2001) and social media (Blankespoor, Miller, and White, 2014; Prokofieva, 2015; Zhou, Lei, Wang, Fan, and Wang 2015) to communicate financial (and nonfinancial) information to users. This provides both a risk and an opportunity to assurance providers. From a risk perspective, auditors are exposed to an increased risk that they will become inappropriately associated with information disseminated on these technology platforms — a risk against which audit firms must safeguard. Beyond assurance services related to privacy and information security (e.g., Toy and Hay 2015), which are escalating in importance, given the frequency and severity of corporate data breaches, social media technology introduces a new demand for assurance. Within social media, credible news reports and sources are intermingled with both biased news and “fake news” — a distinction which is difficult for users to ascertain. Given the media’s role in facilitating the use of information (Drake, Guest, and Twedt, 2014), assurance providers have an opportunity to disentangle biased/fake financial (and nonfinancial) information from credible financial (and nonfinancial) information in social media platforms. This disentangling is important because financial disclosures on personal social media pages and comments thereon have been found to impact users’ perceptions and reactions (Trinkle, Crossler, and Belanger, 2015). Clearly, there are challenges with providing assurance on “real time assertions”; however, auditor JDM research is well equipped to examine the opportunities and risks of a potential future assurance service.

Planning Judgments

Auditing planning decisions include, but are not limited to, audit tasks pertaining to risk assessments, analytical procedures, and materiality — each of these auditing tasks are identified explicitly in Nelson and Tan’s (2005) review. In one of their four audit tasks reviewed, Nelson and Tan (2005) summarized a broad range of literature focusing on auditor risk assessments and audit planning. Based on their review, the literature was mixed as to whether the multiplicative form of the audit risk model adequately captures the auditor risk assessment process, especially, in the presence of fraud. Further mixed

results were identified in how auditors incorporate risk assessments into other audit planning judgments (e.g., resource allocation decisions). Expression of audit risks in terms of quantitative (probabilities) versus qualitative (phrases) was also examined, with results suggesting that auditor risk assessments differed between these terms. Relatedly, task structure (e.g., decomposing auditor risk assessments) was identified as a viable area of future research, given variance in audit firm methodology. Finally, Nelson and Tan (2005) identified that other factors (e.g., audit fee pressures) may moderate audit planning decisions.

In Nelson and Tan's (2005) review, analytical procedures are the second of four audit tasks reviewed. The bulk of the literature examined how factors (e.g., auditor experience, frequency knowledge, client management provided explanations, and incorrect initial explanations) impact auditor hypothesis generation and evaluation. The tendency for auditors to over-rely on inappropriate analytical procedures was also discussed.

Interconnected with the concept of materiality, correction decisions (i.e., book versus waive) is the third of four audit tasks discussed in Nelson and Tan's (2005) review. In general, the research suggested that auditors were more lenient in their book versus waive decisions when auditors were able to justify such treatments or when incentives were structured to favor the client. Perhaps not surprisingly, auditor experience and litigation risk were also found to impact accounting treatment judgments.

Risk Assessments

The literature in risk assessments has expanded greatly over the past decade. To facilitate a review in this auditing task, we have grouped these numerous papers into five overall themes: strategy and business risk assessments, governance risks/effectiveness, risks of material misstatement, fraud risk assessments, and fraud brainstorming.

Strategy and business risk assessments. Auditing standards require auditors to understand "[t]he entity's objectives and strategies, and those related business risks that may result in risks of material misstatement" (CAS 315.11(d)). The literature has questioned whether and how auditors incorporate strategy and business risks into other risk assessments. Kochetova-Kozloski and Messier (2011) study strategic analysis in an auditing context and find that auditors that perform a strategic analysis do not identify more business risks and financial statement risks than auditors that do not perform a strategic analysis, however, the strategic analysis does enable auditors to assess the risk of material misstatement more consistently. Knechel, Salterio, and Kochetova-Kozloski (2010) also investigate the benefits of strategic analysis and find that a strategic analysis enhances the auditors' mental model of the client. This enhanced mental model permits auditors to incorporate greater amounts of information, yielding more balanced and accurate risk assessments (client business risks and risk of material misstatement).

Switching to a business process analysis approach, Kochetova-Kozloski, Kozloski, and Messier (2013) document several benefits from this approach. For example, business process analysis yields a positive association between process-level business risk identification

and business risk identification at the entity level and higher process-level risk of material misstatement assessments. Wright (2016) also finds that for critical business processes, process-specific business risk assessments and risks of material misstatements assessments are positively associated. Further, Wright (2016) modeled the judgment process of entry level auditors, such that at the business process level, the impact of performance on risk of material misstatement assessments is mediated by business risk assessments. However, the literature has highlighted that both auditor training and decision support systems impact auditors' ability to impound business risk information into risk of material misstatement assessments. Specifically, Schultz, Bierstaker, and O'Donnell (2010) find that strategic-systems trained auditors (as compared to transaction-focused) using a decision support system that organized information consistent with the strategic-systems approach are more effective in linking business risk assessments to risk of material misstatement assessments. Moving beyond the linkages to risk of material misstatements assessments, when management provides a fraudulent explanation for fluctuations identified in analytical procedures tasks, Wright and Berger (2011) find that organizing strategic and business process evidence in a business model format (as compared to chronological) enables auditors to provide more precise expectations of nonfraudulent account balances because of improved nonfinancial information cue usage and improved fraud risk assessments.

Governance risks/effectiveness. Auditors obtain an understanding of the control environment (CAS 315.14), including governance (CAS 315.A77-A78), to identify and assess risks of material misstatement (CAS 315.25–26). The literature has highlighted how auditors impound governance effectiveness into audit judgments. In regard to planning decisions, Cohen, Krishnamoorthy, and Wright (2007) suggest that boards of directors differ in their roles of either monitoring management and/or assisting the organization to access resources. They find that auditors impound board role information and strength into control risk assessments and planned audit effort. For example, their results indicate that if the board was strong in either of the above roles, then auditors decrease both control risk and planned audit effort (i.e., hours) compared to if the board was weak in either of the above roles.

The influence of effective governance does not impact only planning decisions. Cohen, Gaynor, Krishnamoorthy, and Wright (2011) examine the effects of audit committee independence from the CEO (e.g., social ties between the CEO and audit committee members) when auditors are waiving proposed audit adjustments. Not surprisingly, in settings where there is a low incentive for managers to engage in earnings management, auditors waive larger proposed audit adjustments, as compared to settings where there is a high incentive for managers to engage in earnings management. However, in settings where there is a high incentive for managers to engage in earnings management, auditors are influenced by audit committee independence. Cohen, Gaynor, Krishnamoorthy, and Wright (2011) explain that this finding is as a result of auditors believing that relatively more independent audit committees improve the bargaining/negotiation power of auditors, thus decreasing the waived adjustments.

Risks of material misstatement. CAS 315.25 requires auditors to identify and assess risks of material misstatement at both the assertion and financial statement level.

In addition to how strategic and business risks impact risk of material misstatement assessments as discussed above, numerous other studies have documented a wide variety of factors that impact these judgments ranging from inputs (e.g., management framing of controls) to outputs (e.g., auditor documentation) of the judgment process.

O'Donnell and Prather-Kinsey (2010) find that auditors conducting analytical procedures from the same auditing firm in the United States, the United Kingdom, and France assess risk of misstatement at the account level differently, yet arrive at similar overall risk of material misstatement conclusions. Specifically, French auditors assess the cost of sales risk of misstatement differently, illustrating nationality differences in audit planning within the same audit firm.

How management assertions are framed to auditors has been found to impact risk of material misstatement assessments. In an accounts receivable task, Fukukawa and Mock (2011) manipulate whether the management assertion is stated in a positive or negative frame (e.g., accounts receivable exists vs. does not exist) and find that the negative frame increases risk assessments, but the extent of this effect is moderated by how and when auditors evaluate evidence (e.g., whether auditors use a probability vs. belief risk assessment approach).

The documentation of risk assessments in auditing working papers has also been examined. Piercey (2011) questions whether the increased focus on documentation by audit regulators may have unintended consequences on auditors' behavior. In a context where auditors are pressured to achieve lenient risk assessments, Piercey finds that additional documentation requirements do not impact quantitative risk assessments (i.e., percentages), but do impact qualitative risk assessments (e.g., very likely). Specifically, additional documentation requirements cause auditors to rationalize their relatively more lenient qualitative risk assessments via word-smithing.

Fraud assessments. The auditors' responsibility for fraud, including identification, assessment, evidence, risk response, and communication, is addressed by CAS 240. Focusing on assessments, auditors are required to assess the risk of material misstatement because of fraud at both the financial statement and assertion level (CAS 240.26). Fraud is characterized by intentional misstatements (CAS 240.2), and accordingly, the auditor is required to treat fraud risks differently (CAS 240.28).

One stream in the fraud literature examines how experience and expertise influence fraud risk assessments. Hammersley, Johnstone, and Kadous (2011) find that audit seniors increase their fraud risk assessments and willingness to consult with a risk management partner when internal control testing results identify a material weakness in an unrelated accounting cycle versus when internal control testing information is not available. Surprisingly, in the presence of materially weak internal controls, auditors create lower-quality fraud detection programs and less efficient audit programs, compared to auditors without control information. Auditors in the material weakness condition tailor their procedures primarily through increasing sample sizes, despite an expert panel indicating that such modifications would not be effective. Recognizing audit quality

concerns, Gold, Knechel, and Wallage (2012) examine fraud consultations in greater detail. Their results suggest that, in contexts of high fraud risk, auditors are more willing to consult with a fraud expert when such consultations are strict (operationalized as mandatory consultation with binding advice and further additional testing would be reviewed by the expert) as compared to when consultations are lenient (operationalized as optional consultation with nonbinding advice and no further review of work by the expert). Gold et al. (2012) also examine how time pressure impacts the willingness to consult an expert. Their mixed results are interpreted by the authors to suggest that missing a deadline is more justifiable in the auditors' mind when there is a heightened risk of fraud. Finally, examining how fraud specialists' judgments differ from auditor judgments in an elevated fraud risk setting, Boritz, Kochetova-Kozloski, and Robinson (2015) document that fraud specialists and auditors select an equivalent number of standard audit procedures with equivalent effectiveness from a standard audit program. However, beyond the standard audit program, the fraud specialists propose more nonstandard procedures that were marginally more effective, albeit less efficient, than the auditors' proposals.

A second stream in the fraud literature examines how client attributes impact fraud risk assessments. Johnson, Kuhn, Apostolou, and Hassell (2013) find that client management narcissism and management's motivation to commit fraud are both positively associated with auditors' fraud risk assessments. Trotman and Wright (2012) examine how external evidence that disconfirms management-controlled internal evidence regarding business process performance is incorporated into auditors' fraud risk assessments. Their results suggest that if internal evidence is consistent (inconsistent) regarding low fraud risk, then auditors do not (do) impound the disconfirming external evidence into their fraud assessments.

Another stream in the fraud literature examines how documentation of fraud planning impacts fraud risk assessments. Examining response modes, Kochetova-Kozloski, Messier, and Eilifsen (2011) find that auditors' judgments are closer to a Bayes' Theorem benchmark when auditors assess fraud cases using frequency responses, rather than probabilities, especially when base rates are low. Favere-Marchesi (2013) finds that when auditors decompose their fraud risk assessments into components (incentives, opportunities and attitude) prior to their overall fraud assessment, they achieve more appropriate overall and component fraud risk assessments, as compared to auditors that categorize fraud risk factors into the components before the assessment and then later assess the components. In response to changes in incentive/opportunity risks, auditors that decompose their judgments also are more willing to revise audit plans and extent of testing than auditors that categorize risks. Hammersley, Bamber, and Carpenter (2010) find that a control condition (i.e., non-primed) of auditors reading a specific fraud risk memo increases fraud risk assessment and evidence requests compared to auditors using a summary fraud risk memo. However, after a time delay, priming auditors to consider fraud risks interact with the documentation condition. Specifically, primed auditors using a summary (specific) memo increase (decrease) fraud risk assessments and evidence requests compared to the non-primed summary (specific) memo auditors. Accordingly, the authors caution that priming may have beneficial effects when using summary

memos, but, because of changing the perceived typicality of the risks, may undermine the benefits of the specific memo.

Fraud brainstorming. CAS 240.16 requires a discussion among the audit engagement team to address "...how and where the entity's financial statements may be susceptible to material misstatement due to fraud, including how fraud might occur." A body of literature examines how fraud discussions/brainstorming may improve audit effectiveness. In an experimental economics study, Bowlin (2011) finds that auditors shift audit resources from low risk accounts to high risk accounts, consistent with risk-based auditing. However, managers anticipate this change in auditor behavior and therefore engage in overriding more in low risk accounts. When auditors are prompted to predict management's perspective/actions, auditors change their audit resources allocation decisions by allocating more resources to the low risk account and maintain the same level of resources to the high risk account, compared to nonprompted auditors. Similarly, Hamilton (2016) finds that in settings of elevated fraud risk, auditors that take the perspective of management increase their assessments that a misstatement was intentional. In contrast, auditors that did not take the perspective of management did not differ in intentionality assessments, regardless of the level of fraud risk. Consistent evidence is further documented by Simon (2012) who finds that prompting auditors to consider management's goals aids auditors in identifying relevant fraud schemes, compared to non-prompted auditors.

Focusing on an individual as an input in group brainstorming, Chen, Khalifa, and Trotman's (2015) results indicate that individual auditors are able to identify a greater quantity and higher quality of frauds when the brainstorming task is done sequentially (i.e., documenting frauds in the categories of revenue, inventory, noncurrent assets, and management estimates) rather than documenting the frauds simultaneously in response to all areas collectively. However, a caveat with sequential brainstorm documentation is that it decreases fraud risk assessments compared to simultaneous brainstorm documentation. In comparison to individual auditor JDM, Carpenter (2007) investigates how audit team JDM differs in a fraud brainstorming task. Her results indicate that team brainstorming increases the quality of fraud ideas generated and fraud risk assessments as compared to individual brainstorming. Hoffman and Zimbelman (2009) extend this research to investigate how to encourage auditors to modify standard audit procedures in high fraud risk settings. Two interventions are investigated: having auditors brainstorm in groups and having auditors engage in strategic reasoning (i.e., considering how management may anticipate the auditors' procedures). Both interventions are effective in improving auditor modifications to the standard audit program, as determined by an expert panel benchmark. Further, both interventions together did not incrementally improve judgments beyond the use of a single intervention.

Not all audit team brainstorming is of equivalent quality. Examining 179 actual audit engagement brainstorming sessions, Brazel, Carpenter, and Jenkins (2010) document variance in brainstorming quality measured by attendance/communication, brainstorming structure/timing, and engagement team effort. Although the results are mixed as to

whether high-quality brainstorming sessions enhance the association between fraud risk factors (incentives, opportunity, and rationalization) and fraud risk assessments, the results provide clear evidence that high-quality brainstorming sessions enable stronger positive associations between fraud risk assessment and fraud risk responses (i.e., nature, timing and extent of procedures, and staffing), compared to low-quality brainstorming sessions. Trotman, Simnett, and Khalifa (2009) find that providing groups with brainstorming guidelines or by engaging in pre-mortem instructions (i.e., analyzing the fraud as if it happened in the past rather than the future) results in an increase in the quantity and quality of generated potential misstatements because of fraud, compared to a group without such guidelines/instructions. Examining both individuals contributing to a team (i.e., nominal brainstorming) and interactive team brainstorming, Chen, Trotman, and Zhou (2015) find that nominal teams outperform interactive teams by generating more unique fraud risk factors, fraud hypotheses, and quality fraud hypotheses. Interacting teams are hindered by inexperienced auditors that engage in social loafing and less developed fraud mental simulations.

A final stream of literature in fraud brainstorming parallels the interpersonal interaction research in review processes (as will be discussed below under “Evaluating Audit Results”): how does the communication medium impact the brainstorming process? Lynch, Murthy, and Engle (2009) find that although brainstorming effectiveness (i.e., identification of relevant fraud factors) does not differ between interactive and nominal team conditions when conducted electronically, electronic brainstorming yields greater effectiveness than traditional face-to-face brainstorming. Facilitating the brainstorming session through prompts (i.e., fraud triangle components and specific risks), however, increases brainstorming effectiveness, as compared to when no prompts are provided. Lastly, post-brainstorming fraud risk assessments are higher than the initial fraud risk assessments, suggesting that brainstorming sessions may improve audit quality.

Analytical Procedures

Analytical procedures are used in risk assessments (CAS 315.6), at the conclusion of the audit when forming an opinion (CAS 520.6) and may be used as substantive procedures (CAS 520.5). Over the last decade, researchers have focused on the importance of analytical procedures in the detection of errors and fraud in financial statements. Some have indicated that better application of cost-effective analytical procedures may have prevented previous accounting scandals (Trompeter and Wright, 2010). Thus, research over the past decade has focused on two areas: (i) improving our understanding regarding what influences analytical procedures and (ii) how to improve analytical procedures to provide better audit conclusions.

Understanding analytical procedures. With analytical procedures accounting for at least one-third of total audit hours (Trompeter and Wright, 2010) and regulators identifying deficiencies in auditors’ applications of analytical procedures (McDaniel and Simmons, 2007), current research in this area has focused on improving our understanding of

what influences auditors' performance of quality analytical procedures. Stemming from regulators' concerns, McDaniel and Simmons (2007) investigate how account expectation precision is influenced by predictability of the account and the level of aggregation of the account details. They then further investigate how consistent auditors' judgments are with their initial expectations. McDaniel and Simmons (2007) find that, when account balances are less predictable (a balance sheet account), auditors assess lower precision for the account balance compared to when account balances are more predictable (an income statement account). They also find that disaggregated data provides greater precision expectations than aggregated data. When comparing to the assessments, auditors judge the level of assurance consistent with their precision assessment, but their judgments of range of differences between the expected and recorded amounts and the likelihood of misstatement are not consistent with their precision assessments.

In a similar vein, Luippold and Kida (2012) investigate the impact of ambiguity in the initial data provided to auditors on performance of two key components of analytical procedures (hypothesis testing and decision accuracy). They vary the level of ambiguity via data sufficiency and data complexity provided to the auditors. Auditors are first asked to develop hypotheses to explain fluctuations in the data after receiving the varying levels of ambiguous data. They then are provided with a second nonvarying set of information and are asked to determine the error in the data. Luippold and Kida's (2012) results indicate that when auditors are initially exposed to more ambiguous information, they do not perform as well at identifying the errors in the financial statements in both their hypotheses and final conclusions, as compared to those that are initially exposed to unambiguous data.

The next set of studies investigate the influence of personal characteristics on the performance of analytical procedures. Cianci and Bierstaker (2009) study the impact of mood (positive versus negative) on auditors' judgments. They find that auditors in a negative mood generate better hypotheses for financial ratio fluctuations, but conversely make less ethical judgments as compared to auditors in a positive mood.

Luippold, Kida, Piercey, and Smith (2015) study management's use of strategic diversion on auditors' abilities to uncover managed earnings. They seed an intentional error to allow management to meet their earnings target and then manipulate whether management diverts the auditors' attention to other accounts with an error or to clean/non-error accounts. Their results show that auditors are least likely to detect the earnings management when they are diverted to the clean accounts and most likely when diverted to accounts with other errors.

Overall, the work in the area of understanding analytical procedures during this period has for the most part supported the regulators' concerns regarding deficiencies in analytical procedures.

Improving analytical procedures. In response to the regulators' concerns and calls for improved audit quality, this stream of research has investigated potential training or aids to improve auditors' analytical procedures performance and judgments.

Moreno, Bhattacharjee, and Brandon (2007) design and test a variety of comprehensive training techniques to improve performance and overcome previously identified deficiencies in analytical procedures. It is expected that training may have differing impacts on the two interdependent processes in analytical procedures (hypothesis generation and hypothesis evaluation). They develop and test three training techniques, “worked-out examples” (learning by reviewing solved problems), “problem solving” (learning by doing the problem), and “self-explanation” (learners actively explaining the solution to themselves). Seven different groups are tested (no training, worked-out example, problem solving, self-explanation, worked-out example with self-explanation, problem solving with self-explanation, and outcome feedback). The study finds that the participants using the combined training techniques (worked-out example with self-explanation and the problem-solving training with self-explanation) perform as well as experienced auditors and better than the other groups. These results differ from prior research, where individual training techniques have improved performance. Further analysis indicates that the individual techniques are effective in hypothesis generation but do not improve the performance in both the generation and evaluation stages.

O'Donnell and Perkins (2011) focus on the diagnostic patterns in the financial statement accounts and study whether the use of a diagram designed around systems-thinking influences auditors to focus more on patterns of change than the use of a diagram designed around business processes. The results show that there is a difference in assessments and judgments for the two diagrams. The systems-thinking diagram auditors make different conclusions about the potential for misstatement and are influenced more by the patterns of the accounts with the seeded errors.

Brewster (2011) also studies the impact of systems thinking on analytical procedures and finds support for improved performance when a systems view is taken. Specifically, he finds that asking participants to take a systems view, versus a reductionist view, when gathering initial evidence, analyzing management representations and evaluating new, incoming evidence, results in improved analytical procedures performance.

Plumlee, Rixom, and Rosman (2015) are also interested in improving performance of analytical procedures and focus their study on metacognitive skills (divergent and convergent thinking). They have three conditions, divergent and convergent, divergent only, and a control and ask auditors to generate explanations for an unusual situation. They find, similar to Moreno et al. (2007), that a combined technique provides a greater likelihood of the correct result. Divergent thinking only increases the number and quality of explanations, however, the combination of divergent and convergent thinking results in the auditors generating and choosing the correct explanation.

Materiality

CAS 320 addresses how auditors determine, apply, revise, and document materiality. Notably, the auditing standards explicitly state that “determining materiality involves the exercise of professional judgment” (CAS 320.A4), highlighting the potential for JDM research in this audit task. However, research on materiality during the time period has

been sparse. The papers reviewed tended to focus on methods or aids that would influence materiality assessments, with a focus on influencing conservative materiality assessments. DeZoort, Harrison, and Taylor (2006) vary auditor accountability pressure levels (review, justification, feedback) to determine how the varying levels influence materiality judgment conservatism, variance, and effort. As the authors predict, the auditors under higher levels of pressure provide more conservative and less variable materiality assessments. These participants also take more time to make their assessments. In addition, the paper finds that when a decision aid was added, the materiality assessments are more conservative and less variable than when a decision aid is not present.

Ng (2007) investigates the influence of earnings thresholds that are potential earnings management targets on auditors' decisions to book or waive audit differences. Results show that when materiality guidance is available, the auditors are more likely to book the adjustments that cause the client not to meet analysts' expectations and not to meet recent earnings, however the guidance does not influence the booking of adjustments that affect a positive earnings threshold. And when the adjustments to be booked are immaterial, auditors have a higher propensity to book the adjustment than when it is material.

Ng and Tan (2007) also study auditor book or waive decisions. They ask auditors to decide whether to book or waive a quantitatively immaterial audit difference that affects the client's ability to meet the analysts' forecast. They manipulate a qualitative materiality factor and the client's concerns about the adverse consequences of missing analysts' forecast. Their results indicate that the auditors are more likely to book the audit difference when the qualitative factor is made salient and the qualitative materiality threshold is lower, but this conservatism disappears when the clients express their concerns about meeting analysts' forecasts.

Summary of Planning Judgments and Suggestions for Future Research

Audit planning judgments have continued to be a thriving area of research given that planning decisions have an influential effect on subsequent audit tasks. Notably, two areas in particular have expanded greatly from Nelson and Tan's (2005) review: strategy/business risk assessments and fraud assessments/brainstorming. Research in these areas have been encouraged by evolving auditing methodology (e.g., Peecher, Schwartz, and Solomon (2007) and auditing standards (e.g., the American Institute of Certified Public Accountants Statement on Auditing Standards No. 99: Consideration of Fraud in a Financial Statement Audit issued in 2002), respectively.

In many of the studies discussed above, it is unclear how risk assessments are conducted within the auditing firms. For example, assessment of the risk of material misstatement occurs at both the assertion and financial statement level, yet the measures used to elicit such judgments tend to lack such specification, calling into question whether auditors are responding in the experimental materials as they would using familiar measures, which may also differ across firms. Deficiencies in analytical procedures continue to be prevalent in auditing, thus future research should further study how auditors apply and interpret analytical procedures. In addition, data analytics and big data

auditing are impacting the application of analytical procedures. The ability to gain a better understanding of financial transactions via data analytics has the potential to significantly change how the auditor analyzes the financial statements of a company and identifies areas of risk. Unusual trends and transactions may be more easily identified; however, clients may be more easily able to smooth trends and hide transactions. A better understanding of how data analytics can be applied and interpreted is necessary. The auditor's ability to analyze all transactions may also change materiality assessments, potentially placing more importance on the qualitative factors to be considered. Future research may investigate additional qualitative factors that might be considered when auditing in a data analytics environment.

Execution and Evaluation Judgments

Although not specifically identified as an auditing task, as discussed in Nelson and Tan (2005), characteristics of the auditor, specialized industry experience and expertise started to be examined by the literature in the late 1990s and early 2000s, with most studies documenting a favorable effect of industry-specialized knowledge. Furthermore, Nelson and Tan (2005) reviewed how decision aids may serve as an efficient means to assist auditors by overcoming deficiencies in performance. Their review highlighted that auditors may excel at issue identification, however mechanical aggregation of information may be superior to auditor judgment. Auditors' confidence in the decision aid is critical as auditors are likely to not rely on a decision aid when they lack confidence in the decision aid or if they are more confident in their own independent judgments. In addition, using decision aids may have deleterious effects such as: (i) novices may not acquire knowledge when relying on such aids and (ii) auditors may game the decision aid by inputting values to arrive at the desired outcome (e.g., sample sizes).

Again, although not explicitly identified as an auditing task, in reviewing between-auditor interpersonal interactions, Nelson and Tan (2005) captured how the literature in the review process grew in the 1990s. From the reviewer perspective, previous research has shown that reviewers attend to inconsistent evidence and that experienced reviewers focus on conceptual issues rather than mechanical issues. From a preparer perspective, the effects of accountability have dominated the literature with a broad range of findings. For example, preparers held accountable became more conservative in their JDM and use a greater quantity of justifications to support their conclusions.

The Use of Specialists and Experts

Auditors are required to consider the competence of the engagement team and the inclusion of experts (CAS 300.2). Research in this area has investigated the effects of having auditors specialize in different industry and audit functions.

Moroney (2007) builds on prior research in this area by investigating the efficiency of auditing specialists when working in specialization versus not in specialization, using two groups of specialists (manufacturing and pension fund). The author breaks down the audit process into three stages to determine if specialist efficiency is similar across

different specializations. She confirms that working in specialization improves efficiency, but the type of specialization effects efficiency at different stages of the audit. The pension fund auditors are more efficient at all stages of the audit, while the manufacturing auditors are less efficient in some aspects of the audit process. Interestingly, both groups provide higher quality decisions when they are less efficient.

In a follow-up paper, Moroney and Carey (2011) study the relative influence of industry and task-based experience on auditors' performance. Past work has shown that both industry and task-based experience can improve auditors' performance over time. Using nonspecialist auditors with various levels of experience in tasks and industries, the authors find that industry-based experience has a greater influence on auditors' performance than task-based experience.

Hammersley (2006) studies the impact of specialization on auditors' abilities to identify misstatements when receiving differing cue patterns. Using specialists matched and mismatched with their specialized industry, the study finds, as expected, that matched specialized auditors are better able to identify and fill-in gaps in evidence cue patterns than the mismatched auditors, resulting in higher quality audit results.

Knechel and Leiby (2016) study the impacts of auditors acting as consultants on audit estimates, with a goal of improving the quality of audit estimates. They manipulate the level of status motives, decision authority and specialized knowledge of the auditor consultants. The participants are asked to provide advice on the client's estimate. Knechel and Leiby (2016) measure the level of contrariness (level of difference with the original estimate) and precision (range for estimate). The results show that status motives and higher specialized knowledge lead to estimate recommendations that are less contrary, but more precise. The precision of the range, however, is tightened in a way that does not counter management's preferred estimate, and therefore, does not necessarily improve audit quality. Higher levels of decision authority result in constrained precision, again potentially negatively impacting audit quality.

Brazel and Agoglia (2007) investigate the influence of accounting information systems (AIS) expertise on audit effectiveness, specifically in the area of risk assessments and subsequent testing decisions and when using a computer assurance specialist. The goal of the article is to examine the relationship between the computer assurance specialist and the auditor and the consequences on the audit. The authors find that auditors with high AIS expertise are more likely to identify and react to potential AIS-specific risks when the competency of the computer assurance specialist is deficient and are more likely to rely on the computer assurance specialist when the competency of the computer assurance specialist is high.

Evaluation of the Internal Auditor

CAS 610 details the requirements for using the work of IAs. The majority of papers in this section relate to how the sourcing arrangements affect the reliance on the IA by the external auditors.

In the first paper, Glover et al. (2008) focus on how alternative internal audit sourcing arrangements affect external auditors' decisions to rely on the work of IAs in the context of different levels of inherent risk and task subjectivity. Results show that sourcing arrangements do not affect planned reliance when inherent risk is low, but when inherent risk is high, external auditors are significantly more likely to rely on outsourced IAs. The authors also find that this reliance on outsourced IAs in the face of increased subjectivity or inherent risk is as a result of external auditors' assessment of the IAs' objectivity, with outsourced IAs being assessed as more objective than in-house IAs.

Brandon (2010) builds on this finding by focusing solely on outsourced IAs. External auditor participants evaluate the competence and objectivity of an outsourced IA where IAs provide no additional services, provide consulting services using the same personnel, or provide consulting services using different personnel. Participants also indicate the extent to which they would rely on the work of outsourced IAs and provide recommended changes to audit hours and fees. Results indicate that evaluations of objectivity are negatively affected by the provision of non-audit services and auditors recommend significant audit fee increases when consulting work is provided, regardless of the outsourcing arrangement.

The third paper on this theme focuses on how the sourcing arrangements of the internal audit function affect external auditors' perceptions of quality, reliance, objectivity, and competence (and therefore reliance) on the IA (Desai, Gerard, and Tripathy, 2011). In addition, the authors also study how additional services (tax) provided by the IA affect the reliance of the external auditors on the IAs. Results show that participants assess the quality and extent of reliance on the outsourced and co-sourced IAs as significantly higher than that of an in-house IA function.³ The planned external audit effort is significantly lower than the planned external audit effort for the in-house IA function, with no significant differences in these measures between the co-sourced and outsourced IA functions. When additional services are added in, the perceived quality of and the extent of reliance on the IA reduces and the planned external audit effort increases. Thus, the general consensus is that external auditors perceive the objectivity of outsourced IAs to be at a higher level than in-house IAs, and therefore, place more reliance on their work. However, as out-sourced IAs perform additional services, this perceived objectivity is impaired and the reliance on the IAs' work diminishes.

Pike, Chui, Martin, and Olvera (2016) seek to understand the coordination between internal and external auditors. The authors find that external auditors' involvement in the review of the IA's plan increases the perceived objectivity of the IA, which leads to greater reported reliance on the IA's work, and a greater reduction of the audit budget. This greater reliance persists, even after a negative audit discovery, compared to external auditors without involvement of the IA, revealing the existence of an anchoring bias. Thus, the authors also identify an unintended consequence of external auditors collaborating with the IAs, which could potentially have an adverse effect on audit quality.

3. Co-sourced IAs exist where there is a partnership between an in-house IA function and an independent IA service provider.

Evaluating Audit Results

JDM research in this area covers an array of different topics and accordingly a variety of auditing standards are pertinent including, but not limited to, standards on how auditors evaluate misstatements identified during the audit (CAS 450) and the appropriateness and sufficiency of audit evidence (CAS 500). The papers explore judgments made during the audit and factors that may affect these judgements. We have grouped these numerous papers into six overall themes: decisions relating to control testing modifications; decision aids and mindsets; moods and emotions impacting decisions; client actions and interactions affecting decisions; firm-interactions affecting decisions; and firm-interactions in the form of review processes.

Control testing modifications. There are two papers that investigate the alternate procedures auditors take in the face of control deficiencies. The first paper investigates whether halo effects from overall risk information and from an unrelated weakness in a compensating control affect evaluations of the effectiveness of a client's compensating control in their internal control over financial reporting (Gramling et al., 2010). Results show that knowledge about a material weakness unrelated to the compensating control to be tested is associated with a preference for a higher level of precision and increased auditor testing, however, results show no evidence of a halo effect because of overall inherent risk influencing these judgements.

Mauldin and Wolfe (2014) attempt to shed light on the underlying reasons why auditors often fail to appropriately modify substantive tests when controls are discovered to be deficient by examining how auditors map a control deficiency into modifications of substantive tests. Results show that a significant minority (33 percent) of auditors incorrectly compensate for a specified control deficiency by increasing sample size. In a follow-up study in the same paper, results show that auditors often make inefficient or ineffective alternative test choices depending on the source of omission caused by the control deficiency.

Decision aids and mindsets. Griffith, Hammersley, Kadous, and Young (2015) investigate how to improve auditors' judgments by increasing critical thinking through a "deliberative mindset" intervention. Mindsets are the set of judgment criteria and cognitive processes and procedures that produce a disposition or readiness to respond in a certain manner (Gollwitzer, 1990), and these mindsets are not dependent on a particular task, but are more global in nature. This article examines whether changing auditors' mindsets can improve audits of estimates, thus improving audit quality. Results show that putting auditors in a deliberative mindset (one which orients users to a careful balanced analysis of alternatives) causes them to assess the client's biased fair value as less reasonable and to choose an audit step that reflects this more urgent concern that fair value is unreasonable.

Desai and Gerard (2013) focus on a more specific treatment rather than a more global mindset. The authors examine whether auditors' consideration of material items, as evidenced by recognition memories, is influenced by income increasing or decreasing

material items. They also investigate whether prompting auditors to form expectations about the possible impact of current material items on future earning affects their consideration of income-increasing and income-decreasing items during the review of working papers. Results indicate that the auditors appear to be more conservative and risk-averse with respect to income-increasing compared to income-decreasing items and have better memories and require more verification for income-increasing items. However, when prompted to form expectations about the future impact of these items, there was no difference in the recognition memory or confidence in memory of the material income-increasing or income-decreasing items.

Backof, Bamber, and Carpenter (2016) investigate how different judgment frameworks affect auditors' constraint of management's aggressive financial reporting under accounting standards that differ in their level of precision. The authors find that judgment frameworks are more effective when accounting standards are less precise. Specifically, a framework that requires auditors to consider the pros and cons for reasonable alternatives effectively enhances auditors' skepticism of aggressive reporting under less precise standards. A further reduction in allowance of aggressive reporting is found when variations of a structured thought process is also included with the framework (i.e., answering "why" accounts for the transaction in a certain way).

Brewster (2016) investigates how client lies affect an auditors' evidence-based recollection and whether a decision-aid can assist auditors in resisting these lies. Results show that auditors provided with a decision-aid to help them understand a complex situation better are less susceptible to managements' lies and provide lower evaluations of management credibility. However, the auditors who are not provided the decision aid do not have an in-depth understanding of the initial concept and are more susceptible to client lies. Follow-up tests show that the misinformation effect is not because of improperly gathering evidence and that these auditors are as confident as the other auditors in their prior knowledge and understanding.

Mood and emotion. Mood is a state of mind or feeling that is task-irrelevant, yet it could impact auditor judgment. Chung, Cohen, and Monroe (2008) explore whether auditors in a positive mood condition exhibit lower consensus and less conservatism relative to auditors in a negative mood condition. They find that positive-mood participants' approach to decision-making results in the lowest judgment consensus when compared with the neutral and negative-mood participants. They also find participants in the positive mood condition make less conservative inventory judgements than the negative-mood participants, demonstrating that moods have the potential to affect audit judgment in an ambiguous task setting. Thus, a positive mood in an audit context may be unconstructive because of its effect on conservatism and consensus.

Bhattacharjee, Moreno, and Riley (2012) continue along this theme and study emotions and their impact on auditor judgment. They seek to examine if interpersonal affect affects auditors' sensitivity to their judgment of client competence, which then impacts persuasiveness of evidence. Results show that negative affect of lower competence management has efficiency implications as a result of increased testing being performed.

Positive affect led to the competence effect being eliminated in lower competence management, thus having effectiveness implications for the audit.

Lastly, affect has also been examined in the review process. Frank and Hoffman (2015) question whether a reviewer discounts the preparers' conclusion when it appears to be inconsistent with the working papers. Consistent with prior research, the results suggest that when the reviewers are not aware of the preparers' affect, reviewers discount preparers' conclusions and instead base their judgments on the working paper evidence. Surprisingly, when reviewers are aware of the preparers' affect, the reviewer relies more on the preparers' conclusion highlighting a potential limitation of the review process.

Client actions and interactions. Auditors interact with clients on a regular basis, and these interactions can have important impacts on the process and outcome of the audit. Bennett and Hatfield (2013) study how social interactions between staff-level auditors and client management may affect staff auditors' evaluation of audit evidence. They find that the mismatching of staff auditors on age, experience and accounting knowledge with client management results in staff auditors avoiding interactions with client management when gathering evidence. Results also show that staff auditors' use of email communication mitigates the reduction of evidence gathered, but when evidence is not fully gathered, the documentation is vague and presented in a manner such that more senior auditors would not identify a problem.

The ability of management to persuade auditors' evidence evaluation has been investigated by two studies. Kaplan, O'Donnell, and Arel (2008) question whether experience enables auditors to increase persuasion knowledge and thereby decrease the impact of management's persuasion attempts on auditors. Kaplan et al.'s results document that when management evaluates their controls as weaker than what independent evidence suggests, auditors' experience does not impact auditors' risk assessments (i.e., reliability assessment of internal controls over sales process). However, when management evaluates their controls as stronger than what independent evidence suggests, auditors' experience does impact auditors' risk assessments, such that more experienced auditors are less likely to be influenced by management's assessment. Further, Wolfe, Mauldin, and Diaz (2009) study how management uses persuasion tactics to affect auditors' assessments of deficiencies rising from internal control deviations. The authors study two types of persuasion tactics: concessions and denials, in both IT and manual controls. They hypothesize and find that auditors are more persuaded by concessions and assess the significance of deficiency arising from IT control deviations lower for concessions than denial. There is no difference in assessment because of persuasion tactics in manual controls.

In the audit of financial statements, auditors are given the financial statements, and thus start with the knowledge of what decisions have already been made by management in preparing the financial statements. Thus, management is the "first-mover" in this situation. Earley, Hoffman, and Joe (2008) examine the "first mover" advantage and investigate how awareness of management's severity classification of internal control problems (knowledge bias) influences auditors' initial assessment of internal control over financial reporting. They test a potential solution to this unintentional cognitive effect by

cognitively restructuring the task and show that auditors' judgements are biased by prior knowledge of management's classification. Results also show that cognitively restructuring the task by instructing auditors to explicitly consider and document the potential financial statement impact of internal controls over financial reporting mitigates management's "first mover" influence on auditors' judgements.

Clients can also impact auditor judgments and rebuff auditors' potential concerns by adding supplemental note disclosures. Griffin (2014) examines how two types of uncertainty (subjectivity and imprecision) and supplemental footnote disclosure affect auditors' decisions to require fair value adjustments. Results show that subjectivity and imprecision interact to increase the likelihood that auditors will require adjustment of fair value estimates when the estimate is imprecise and more subjective. However, this interaction is negated when supplemental footnote disclosure is included. Results also show that when there is no disclosure, the dollar amount of adjustment that is required is based on imprecision only and consistent with standards, the lower bound of the range of misstatements is used rather than the midpoint.

Firm interactions. Firms have the ability to aid auditors in making better judgments by altering the process, instructions or interventions surrounding information gathering and analysis they give to members of the firm. Payne and Ramsay (2008) examine the firm's use of different audit documentation methods, summary memos or detailed working papers and the effect of their use on auditor's cognitive processing. They then further investigate the impact of cognitive procession on memory and performance (detection of errors and patterns indicative of a fraud). Results show that different documentation methods lead to auditors' taking different cognitive processing routes to achieve performance, which is measured by level of error detection and recognition of a pattern that may increase fraud. Detailed documentation increases the total amount of time spent on a task and time spent is positively related to both error detection and recognition of fraud patterns. Summary memo documentation results in examining evidence a greater number of times and this increases memory about the evidence and increases pattern recognition.

Rasso's (2015) study builds on these findings and investigates how documentation instruction affects the collection and processing of audit evidence. Results show that instructions that promote high-level construals (interpretations) help auditors abstract information and enable them to see the big picture, resulting in an increased level of professional skepticism and better processing of audit evidence than instructions that promote low-level construals or no instructions. The high-level construal instructions also lead to auditors spending significantly more time on the task, collecting significantly more audit evidence (than the low-level construal instruction group) and rating the fair value estimate as significantly more risky.

Messier, Quick, and Vandervelde (2014) study the use of process accountability within the firm. The authors investigate whether auditors exhibit a status quo heuristic when interpreting accounting standards and whether process accountability will reduce the effect of the status quo heuristic. Results show that auditors are more likely to choose

the prior year treatment when determining the appropriate treatment for the current year when process accountability is low. However, this effect disappears when process accountability is high.

Peecher, Piercey, Rich, and Tubbs (2010) explore how supervisor intervention affects finalization of audit team judgements, when supervisors have may have directional goals and perceive a technical knowledge over subordinates. Results show that subordinates' judgements are influenced by this active intervention and that the subordinates' (influenced) judgements are reflected in supervisors' finalized judgment. Further, this intervention leads to incremental bias in supervisors' finalized judgment beyond that attributable to any effects on subordinates' judgements. In a second experiment in the same paper, results show that audit supervisors do not understand the bias-inflating effects of their intervening in subordinates' judgment.

Ng and Shankar (2010) also study interactions within the firm and investigate the impact of Quality Assessment Standards (QASs) on auditors' decisions to accept client preferred accounting alternatives and how clients' justification and nature of technical departments' recommendations jointly influence these decisions. Results indicate that, without advice from the technical department, auditors' decisions to accept the client-preferred method are significantly influenced by the strength of the client's justification, but not by the presence of a QAS. A significant three-way interaction is found between the presence of a QAS, the nature of the technical departments' advice (explicit recommendation or advice), and client justification. When the client's justification is strong, auditors' propensity to accept the client-preferred alternative decreases in the presence of QAS, but only when the advice from the technical team is an explicit recommendation.

Firm interactions: Review process. The interpersonal interactions in the review process continue to be a fruitful area of research. Payne, Ramsay, and Bamber (2010) question whether anticipating the form of review (real-time interactive versus traditional written review notes) impacts preparer behavior. In an accounts receivable control testing task, the results indicate that performance on the mechanical steps of the audit program (i.e., tracing, inspecting for authorization) are unaffected by the anticipated review process. However, anticipating an interactive review increases the thoroughness of the cognitively demanding steps of the audit program (i.e., reviewing write-offs for unusual items) compared to the traditional review process. Furthermore, path analysis reveals that, although performance on the mechanical steps in the audit program is positively associated with the identification of mechanical errors, performance on the cognitively demanding audit program steps are positively associated with identification of mechanical errors and fraud identification. However, a draw-back of the real-time review process is investigated by Tan and Tan (2008) in which they suggest that in contrast to a traditional review, real-time review exposes reviewers to preliminary evidence that may be subsequently invalidated. They document that the invalidated evidence continues to influence both preparers and reviewers. The effects of the invalidated evidence are mitigated when the reviewers are informed that their judgments will be communicated to the preparers,

however the bias in the preparers' judgment are not mitigated when they are informed that their judgments will be communicated to the reviewers.

Recent research suggests that different methods of review may alter how auditors prepare audit documentation with audit team members perceiving electronic review as less demanding. Agoglia, Hatfield, and Brazel (2009) investigate how the format of audit review (face-to-face or electronic) affects review team judgements. The experiment matched audit seniors (preparers) with audit managers (reviewers) and manipulated review format. Results show the effect of review mode persists through the review process, with electronic reviews resulting in lower quality reviewers' going concern judgements when preparers anticipate an electronic review. Mediation analysis suggests it is the documentation quality assessment gap (reviewer perceptions of working paper documentation quality versus actual documentation quality) that mediates this relationship (vs. the documentation quality issues).

Another set of studies examine how preparers justify an audit conclusion to a reviewer and how reviewers respond to such justifications. Coyne, Biggs, and Rich (2010) investigate the effects of anticipating different review formats (real-time interactive reviews versus traditional review with corresponding review notes not done face-to-face) on preparers' choice of audit procedures. Results show that anticipating an interactive review leads auditors to focus more on cognitively demanding, conclusion-oriented audit procedures, leading to better performance in identifying items that may be fraudulent. Shankar and Tan (2006) document that the method to which auditors justify a conclusion depends on the auditor's knowledge and the preferred conclusion of the reviewer. They show that if the preferred conclusion of the reviewer is different from the conclusion of the preparer, the preparer with high knowledge (both managerial and technical) will consider a broader range of issues and use pro versus con reasoning in the justification. However, if the preferred conclusion of the reviewer is similar to the conclusion of the preparer, a high managerial knowledge preparer uses an evidence framing approach to focus on the consistent evidence in the justification. Tan and Shankar (2010) further this stream of literature by showing how reviewers assess the preparers' work and justifications. Specifically, audit reviewers provide greater performance evaluations to preparers whose justification memos and conclusions are consistent with the reviewer's opinions. However, this relationship is moderated by whether the reviewer believes it is important to have congruent opinions.

As discussed above, the review process has many contextual features that impact its efficiency and effectiveness. Lambert and Agoglia (2011) focus on two further features of the review process, timeliness of the review and the review note framing by the reviewer, to investigate how these contextual features impact reviewee follow-through. Broadly, compared to a timely review, a delayed review elicits lower reviewee effort and increases dysfunctional behavior (i.e., over-documentation, aka ghost ticking). Supplemental analyses provide evidence that the review frame has the potential to interact with the timeliness of review. Specifically, when the review is done in a timely manner, a conclusion-framed review note leads to greater reviewee effort and performance than a documentation-framed review note.

Summary of Execution and Evaluation Judgments and Suggestions for Future Research

Several important advances have occurred in the literature over the past decade in the area of execution and evaluation judgments. First, numerous studies discussed above have answered the call from Nelson and Tan (2005: 54) for research on how auditors' affect/emotions influence JDM. Second, the scope of auditor/client interactions have increased beyond just auditor client-management negotiations as previously documented by Nelson and Tan (2005). Third, research on the review process has changed focus from accountability issues to format/technological differences that may directly impact audit effectiveness and efficiency. Auditor-client negotiations could also be considered a task in this audit process. We did not include this task as part of our literature review as Salterio (2012) recently did a comprehensive review of the auditor negotiation JDM literature, and therefore, there were few papers published in this area subsequent to this review (see Salterio (2012)).

In regard to future research pertaining to execution and evaluation judgments, one such area is how auditors effectively assess management's fair value measurements (e.g., Cannon and Bedard, 2017). As accounting standards embrace inclusion of more relevant information in the financial statements for user decision making, there are challenges in how auditors assess and conclude on the reliability of such information. Second, big data analytics, defined as "the process of inspecting, cleaning, transforming, and modeling Big Data to discover and communicate useful information and patterns, suggest conclusions, and support decision making" (Cao, Chychyla, and Stewart, 2015: 423) may transform how audits are conducted (Alles, 2015; Krahel and Titera, 2015). Although specific future research questions in big data have already been advanced by the literature (Brown-Liburd, Hussein, and Lombardi, 2015), the above definition of big data analytics provides a road map of the many judgments and decisions auditors must make when utilizing big data analytics and relying on big data as audit evidence (Yoon, Hoogduin, and Zhang, 2015); and hence, facets that audit JDM research may assess and improve. These are just two examples of how auditor competencies are evolving and being challenged in the traditional financial statement audit. Auditor JDM research is well suited to address how auditors may develop task expertise rather than passing these tasks to other professions (e.g., specialists).

Reporting Judgments

In Nelson and Tan's (2005) review, going concern/opinion modification judgments is the fourth of four audit tasks reviewed. This stream of research has a long history in the auditing literature, but has declined in recent years as evidenced by a single study in the 2000's focused on the audit task. As discussed in Nelson and Tan, this study documented that auditor opinion modification was found not to be influenced by litigation risk but rather through redundant information increasing the saliency of default risk.

Evaluation of Going Concern

Going concern has always been a fundamental principle in the preparation of financial statements. There has been scant research in this area recently, but with new reporting

requirements and increased attention focused on both managements' and auditors' going concern assessments, we anticipate future research potential in this area. CAS 570.06 details the auditor's responsibilities to obtain sufficient appropriate audit evidence regarding the appropriateness of management's use of the going concern basis of accounting in the preparation of the financial statements, and to conclude, based on the audit evidence obtained, whether a material uncertainty exists about the entity's ability to continue as a going concern.

Bruynseels, Knechel, and Willekens (2013) study how management's turnaround initiative influences auditors' going-concern decisions for financially distressed firms. They hypothesize that the judgment of auditors will be influenced both directly and indirectly when information is available early in an audit engagement where a client suffers from financial distress and is undertaking efforts to mitigate its financial problems. An indirect effect is expected because the presence of company turnaround initiatives is also likely to have an indirect effect on auditors' perception of client viability by changing the focus of the evaluation of financial information that is considered later in the audit. They find that strategic turnaround initiatives have little effect on financial recall or going-concern judgments, suggesting they are not viewed negatively or positively by an auditor. However, turnaround initiatives related to operations, such as, cost-cutting, influence an auditor's going-concern assessment, and that effect arises because the information causes an auditor to reduce his/her focus on subsequent positive financial information.

Guiral, Ruiz, and Rodgers (2011) investigate whether the self-fulfilling prophecy effect (motivated reasoning) results in auditors' reluctance to issue going concern opinions. They find that auditors' expectations of the self-fulfilling prophecy affect their attitudes toward confirming and disconfirming going concern evidence. Auditors who believe in the self-fulfilling prophecy have a greater sensitivity to mitigating evidence that shows viability information and, at the same time, a lower tendency to favor contrary evidence.

The third paper in this theme looks at the ordering of evidence in a going concern decision. Favere-Marchesi (2006) investigates whether auditors take into consideration the chronological order (trend effect) when processing mixed evidence, or if auditors' evaluations are influenced primarily by the order of presentation — (recency effect), as found in many prior studies. Trend effect is very important to consider when evaluating going-concern decisions, but auditors may not always review information in chronological order. The author finds trend effect dominates the recency effect in auditors making going concern decisions, and thus auditors can overcome the recency bias because of their experience as decision makers in going concern decisions.

Summary of Reporting Judgments and Suggestions for Future Research

These three studies have advanced our understanding in the last decade by examining how auditors use evidence, both past and future in their going concern opinions. Specifically, cost-cutting behaviors were viewed more harshly by auditors versus strategic turnaround efforts. Studies also showed that auditors can overcome recency bias in the

evaluation of going concern evidence but are still susceptible to the self-fulfilling prophecy effect.

Despite the paucity of research in recent decades in auditor reporting, this area of research is poised to increase significantly given the recent auditing standard changes pertaining to the audit report. The revised auditing standards on auditor reporting have drastically increased the firm-specific information contained in the audit report to financial statement users. Of particular importance, disclosure of key audit matters (CAS 701) and identification of the engagement audit partner (CAS 700.46) potentially provides useful information for financial statement users. In regard to the former, recent behavioral research has questioned how key audit matter disclosures impact auditor legal liability, by examining juror JDM (e.g., Brasel, Doxey, Grenier, and Reffett, 2016; Gimbar, Hansen, and Ozlanski, 2016). However, it is unclear how auditors decide which key audit matters are to be discussed in the audit report, how to disclose complex information for financial statement users, whether such information is interpretable and useful for financial statement users, and how firm quality control procedures (e.g., legal counsel) may interact with the transparency of the key audit matter disclosures. The expectations gap of financial statement users as it relates to the audit report (e.g., Gray, Turner, Coram, and Mock, 2011; Bedard, Sutton, Arnold, and Phillips, 2012) underscores the difficulty of auditors' JDM processes pertaining to the key audit matter disclosures.

In regard to the latter, identification of the engagement audit partner, prior archival research has differentiated audit partner quality by means of client accrual quality and has suggested that markets respond to disclosures of the audit partner name (Aobdia, Lin, and Petacchi, 2015). Further, archival analyses have documented that audit partner expertise/specializations impact audit opinion modification (Chi and Chin, 2011) and client acceptance decisions (Hsieh and Lin, 2016). However, research has questioned whether the disclosure of the audit partner name increases audit quality *in fact* rather than merely *in appearance* (King, Davis, and Mintchik, 2012). King, Davis, and Mintchik (2012) suggest that the increased accountability/transparency provided by the disclosure may decrease audit efficiency by over-auditing (see also Carcello and Santore, 2015); however, the impact on audit effectiveness is tenuous. Future auditor JDM research will be able to address how disclosure impacts audit quality and financial statement user JDM.

CONCLUSION

The purpose of this manuscript is twofold. First and foremost, the manuscript has endeavored to provide the pulse of auditor JDM research published over the past decade since Nelson and Tan (2005). In doing so, the manuscript has structured the literature review to focus on explicit tasks in the auditing process to facilitate researchers and graduate students in understanding the current JDM research *in auditing*. Readers familiar with the Nelson and Tan (2005) review will readily notice the differences in structure between the two manuscripts. Specifically, whereas Nelson and Tan (2005) decomposed their review to focus on the interdependent areas of auditing task, auditor person and interpersonal interactions, we focused exclusively on the auditing task. Because of this difference, the number of auditing tasks examined in our review have increased

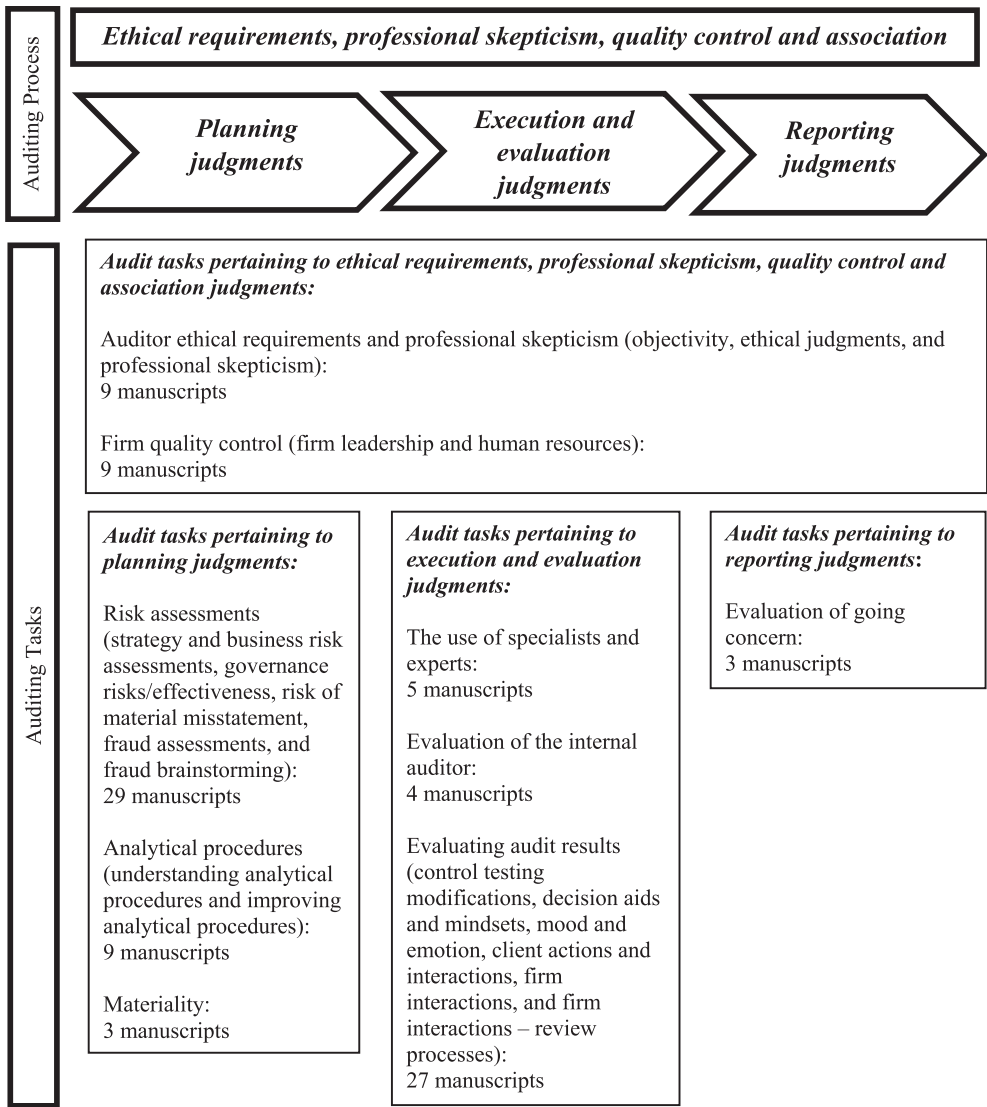
compared to the four tasks examined by Nelson and Tan (2005) (risks, analytical procedures, materiality, and reporting). However, Nelson and Tan's (2005) discussion of the auditor as a person and interpersonal interactions implicitly addresses auditing tasks which we have included in our review. For example, our discussion of ethical requirements, professional skepticism, quality control, and association align closely with Nelson and Tan's (2005) review of the auditor as a person given the subject matter (e.g., expertise). In regard to interpersonal interactions, between-auditor interactions involve review processes which we have identified as an auditing task. In sum, our literature review has not excluded interpersonal and personal aspects from Nelson and Tan (2005), but rather has situated them in the appropriate tasks in the auditing process.

The auditing literature over the past decade has evolved tremendously. Consistent with the calls for future research in Nelson and Tan (2005), the literature has experienced growth in the areas of strategic systems auditing and auditor interactions, such as review processes (see also Salterio, 2012). Auditing tasks pertaining to fraud have attracted immense attention both from an individual perspective (e.g., fraud risk assessments) and a group perspective (e.g., brainstorming activities). The influence of mood and emotions has also gained traction in the literature. On the other hand, despite the call from Nelson and Tan (2005), JDM research in auditing tasks related to internal control and certification requirements in integrated audits has not increased substantially even though auditing standards evolved during this time (albeit related archival research has flourished). Equally surprisingly, fundamental auditing tasks (e.g., client acceptance and continuance decisions, association) have received no attention from the auditors' JDM perspective. Accordingly, despite the many insights achieved over the past decade, much is left for future JDM research to address.

Given the above literature review is restricted to research published within the last decade, identification of future research has been done judiciously given that manuscripts published previously may have already examined the topic and unpublished working papers are currently extending the literature. Accordingly, the manuscript identifies several auditing tasks throughout the audit process that, in the authors' opinion, have the potential to transform the auditing institutional environment. In the ethical requirements, professional skepticism, quality control and association audit process stage, fundamental principles contained in the professional codes of conduct, elements of firm quality control and matters of association are under-investigated, despite their influence on judgments in the auditing process. Financial statement users' demand for information in social media and technologies will increase tension on traditional auditor competencies, which provides avenues for future JDM research to address. In regard to audit planning judgments, how auditors process entity level information (e.g., business strategy, business risk, and governance attributes) and impound such information into their planning (e.g., fraud brainstorming) will continue to be important. In addition, evolving audit tools such as data analytics has the potential to change how auditors assess risk, determine materiality, and interpret analytical procedures. Further strain on auditor competencies were identified for future research in the execution and evaluation stage of the audit, in particular accounting matters such as how auditors address fair value measures and auditing matters

such as the potential for big data analytics. Finally, in the reporting stage of the audit process, changes in auditing standards, in particular reporting on key audit matters, merit future JDM research to understand, explain, and predict how and why auditors address these evolving challenges.

APPENDIX
CATEGORIZATION OF LITERATURE BY AUDIT TASK IN THE EXTERNAL FINANCIAL STATEMENT AUDIT PROCESS.



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